

Plan Overview

- 1) Everyone tries with a very basic prompt initially e.g. “I would like to process some ATAC-Seq data, can you please write me a script to generate bam files mapped to the mouse genome.”
 - Could also just show it the accession numbers we’re interested in and see if it can come up with download script too
 - See what we get and if it runs
- 2) Next step would be to try a more guided and specific prompt (particularly focused on QC, if it hasn’t already included this)
 - Goal would be to end up with some sort of QC reports, there are three I’d normally check FastQC, FastQScreen, and muti-QC
- 3) Then final step would then be to see if the LLM can assess the QC output. We may need to pass it the html/ figures for it to review.
 - The critical concern in this dataset is that there is mycoplasma contamination in one of the conditions
 - So, we want to see if the LLM correctly identifies concerning metrics
 - What advice does it give
 - Does it identify that mycoplasma is a confounding variable between conditions that means it will be very difficult to cleanly interpret the biological meaning behind the results.

We can also repeat these across a few LLMs, if we have time

Recording Feedback

More general questions on user background

- Name/ User ID
- How much experience do you have:
 - With processing sequencing data
 - Coding
 - Using LLMs

Records to keep track of when generating a programming pipeline script:

- What LLM are you using
- What type of prompt did you give (basic/specific)
- What is the first prompt you gave
- Did it tell you to run a QC program(s)
 - If yes which ones
- Does the script run successfully
 - End goal to have mapped bam files for each sample
 - End goal to have QC reports (if instructed)
 - More nuanced might be to ask what options it uses for programs e.g. trimming and mapping – can we determine if they seem reasonable (for standard approach)
- Did you have to troubleshoot with the model e.g. did it require further prompts to get a workable script (if so how many prompts)

Records to keep track of when reviewing the QC results

- What LLM are you using
- What type of prompt did you give (basic/specific)
- What is the first prompt you gave
- Did it flag any concerns with the data
 - If yes what metrics did it highlight
- Did it identify the mycoplasma contamination as a concern
- Did it still think the data was usable
 - If yes were there any cautions (with or without additional prompting)